Principle of Pharmacogenetics

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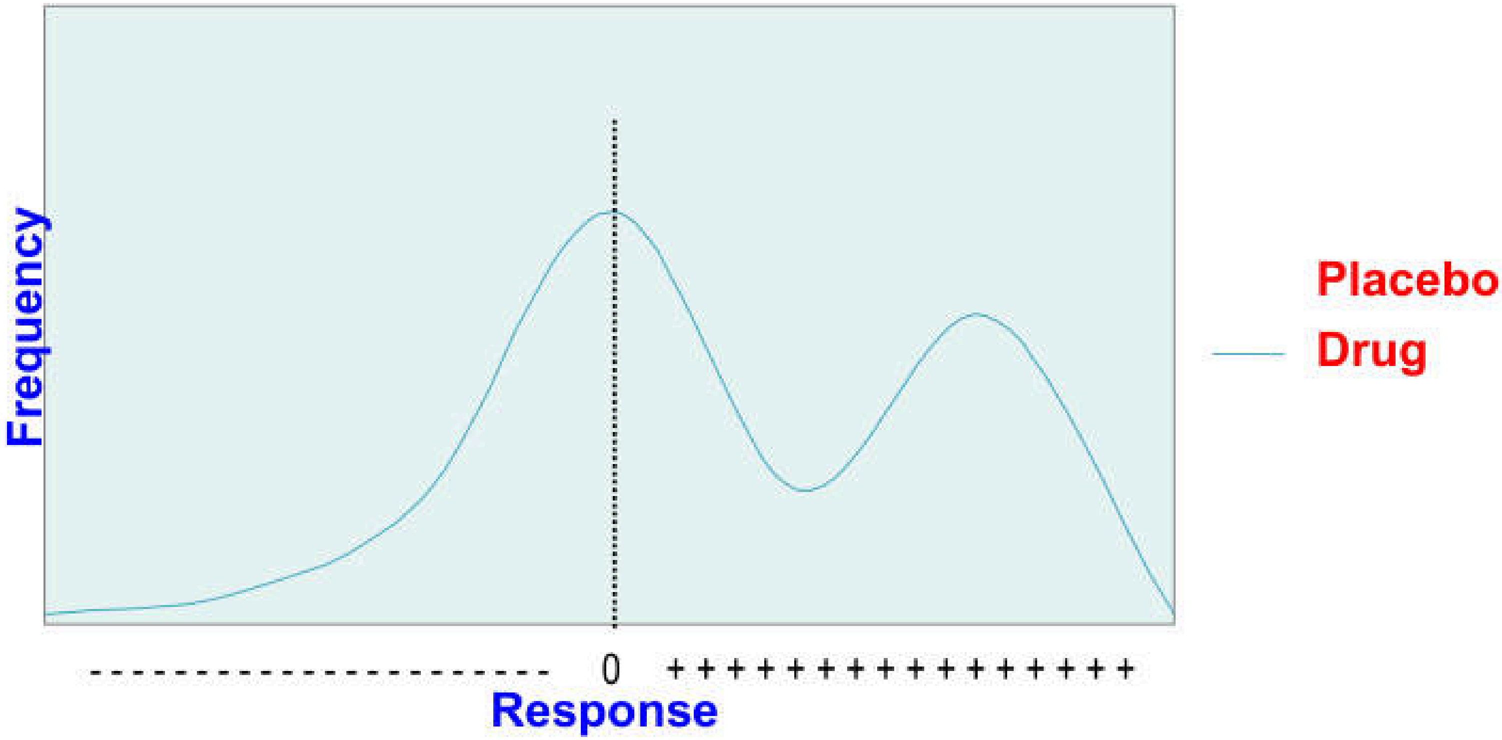
Different Drug Response: Efficacy & Toxicity



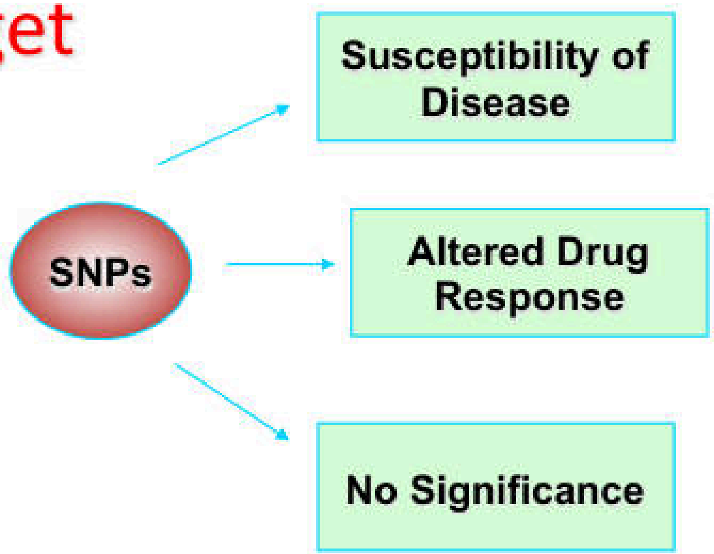
Efficacy = Higher probability of clinical improvement in controlled patient population, compared with placebo, using selected, relevant clinical endpoints

...... The drugs don't work.......

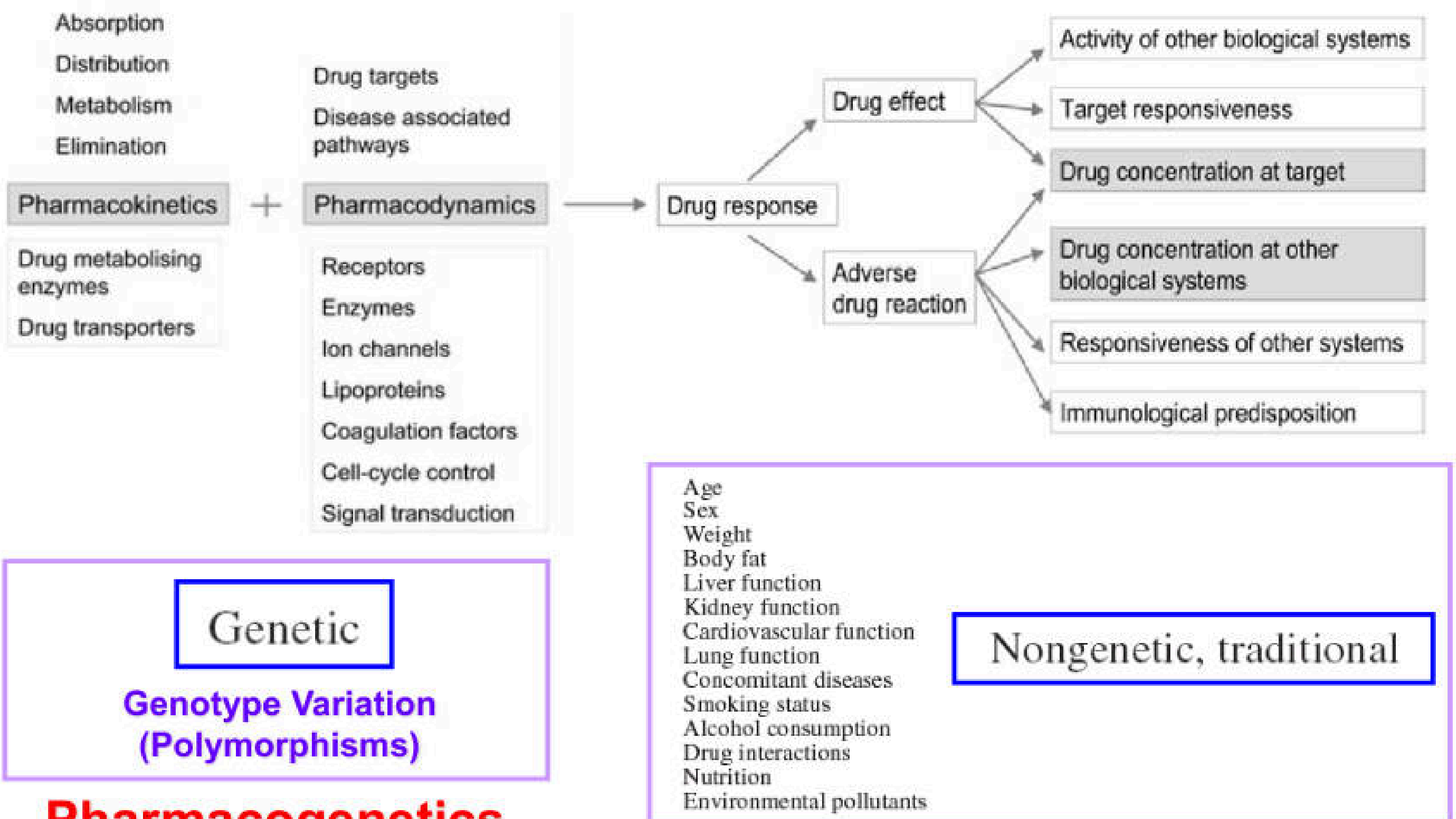
.... They just make it worse.......

Factors Influencing Efficacy and ADR

- Pharmacokinetic properties
  bioavailability-related issue
  - Drug uptake & transport
  - Drug distribution
  - Drug metabolism
  - Drug elimination
- Pharmacodynamic properties
  drug-effect-on-target issue
  - Receptors
  - Enzyme
  - Drug transporters
  - Disease associated pathway
    - Second messenger
    - Effector molecules
- Off-target



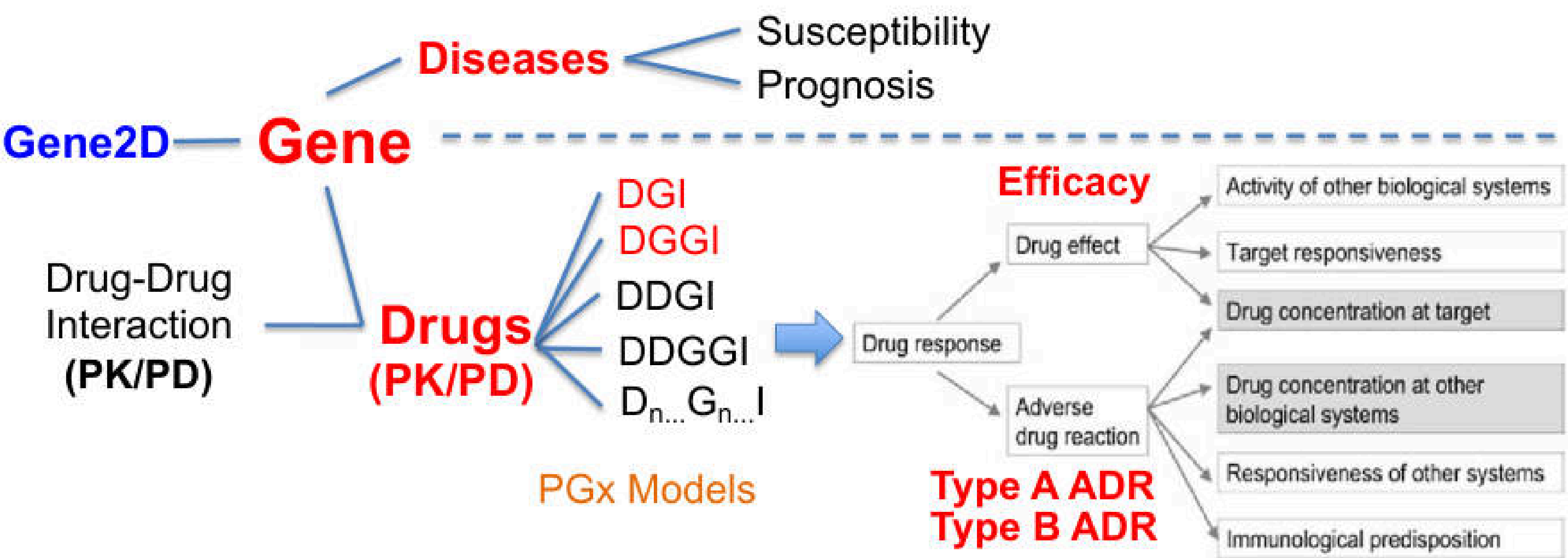
Parameter Determined Individual Variability in Drug Effects and ADR



Pharmacogenetics

J Mol Med (2003); 81:154-167.

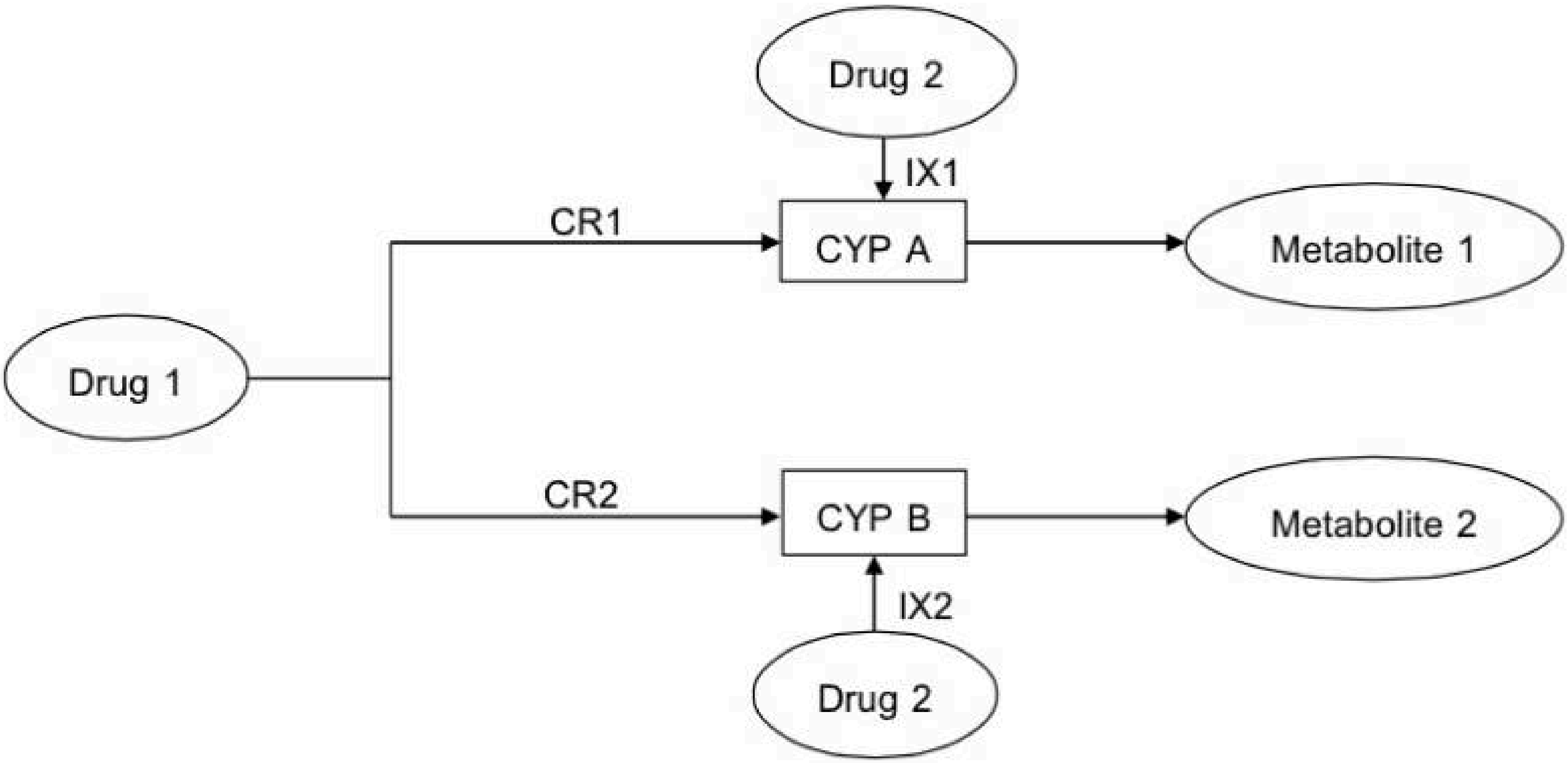
PGx Models:
Variability in Drug Effects and ADR



Pharmacogenetics: The study of the influence of genetics (variations in DNA sequence) on drug response.

DNA → RNA → PROTEIN → PHENOTYPE

PK-Drug-Drug Interaction (DDI) Model

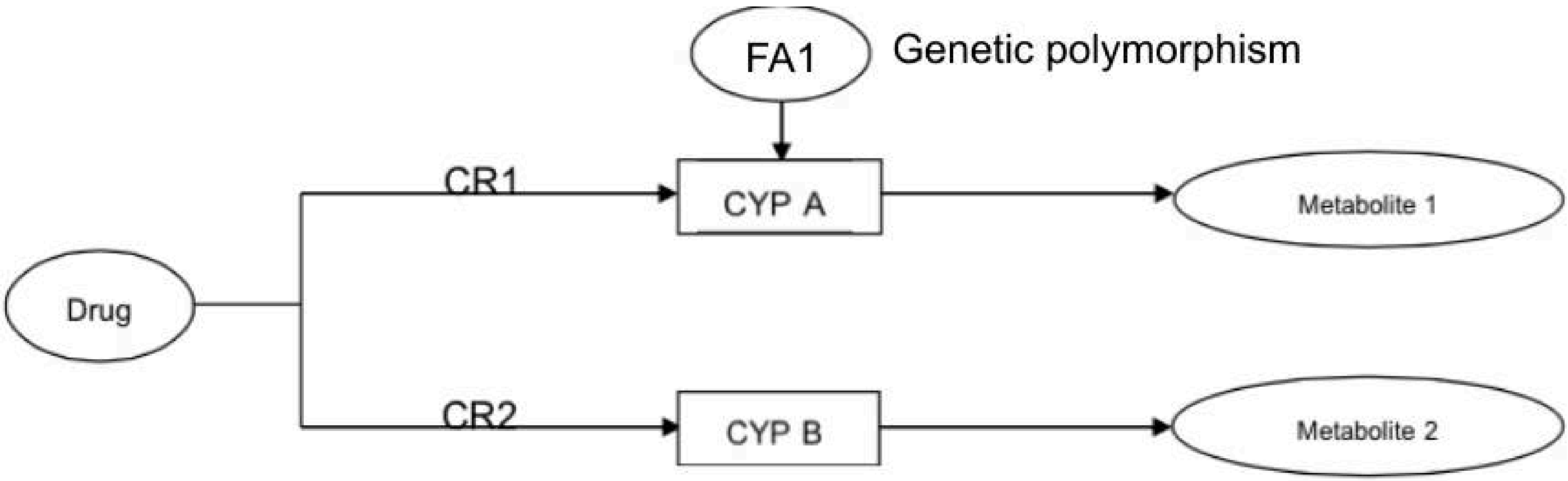


CR1, CR2 = contribution ratio ที่ยาผ่าน CYP A และ B ตามลำดับ โดยผลรวมของ CR1, CR2 ต้องน้อยกว่าหรือเท่ากับ 1

CYP A, B = CYP ที่ Drug 1 ถูกเมแทบอไลต์

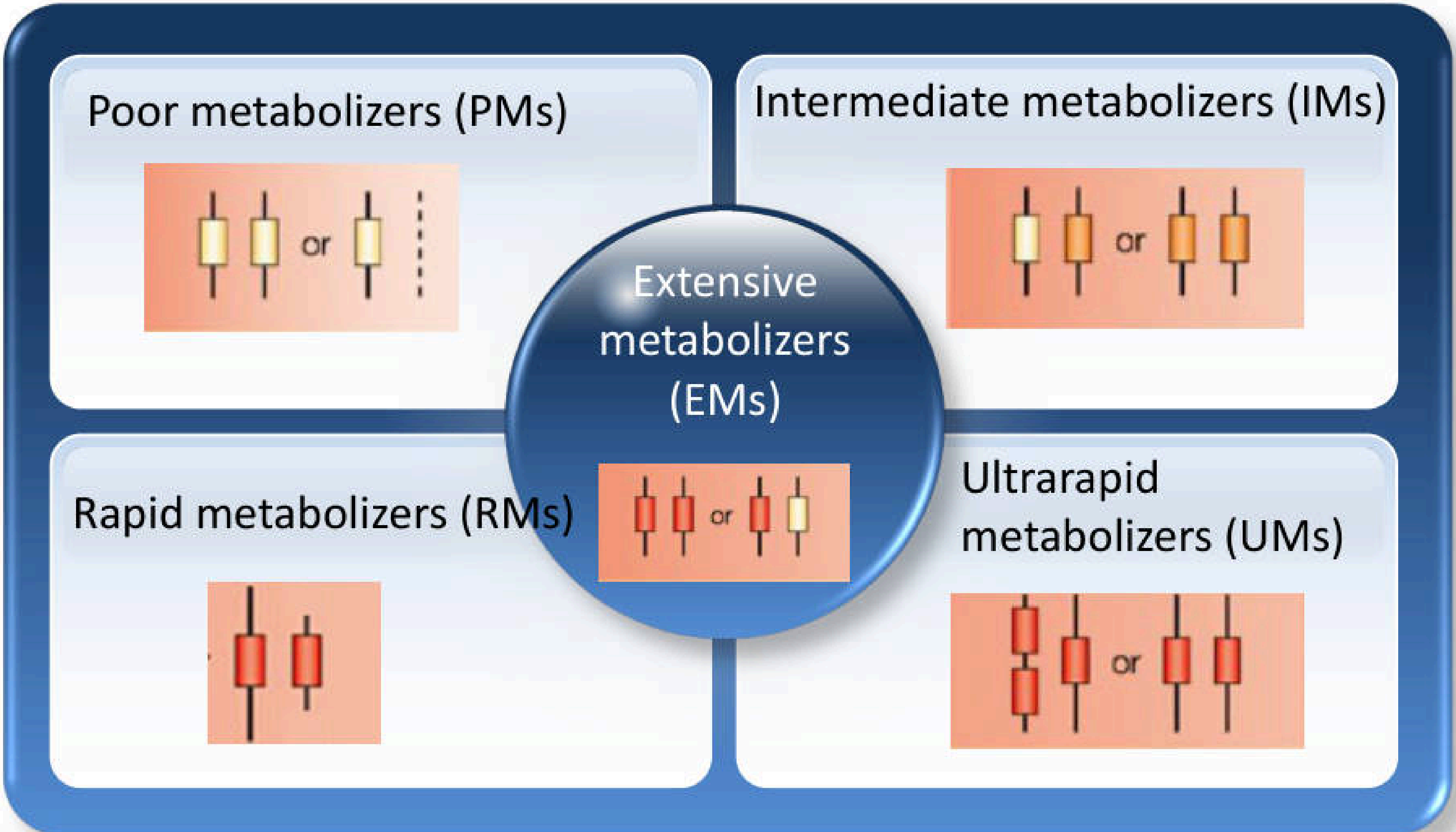
IX1, IX2 = ความแรงของตัวยับยั้งหรือตัวกระตุ้น เป็นความสามารถในการยับยั้งหรือเหนี่ยวนำการทำงานของ CYPs มีค่าตั้งแต่ -1ถึง+1 โดยค่าลบแสดงถึงการยับยั้ง ค่าบวกแสดงถึงการเหนี่ยวนำ ค่าเป็นศูนย์คือไม่มีผลต่อการทำงานของ CYP

PK-Drug-Gene Interaction (DGI) Model



FA = fractional activity คือ ค่าที่บ่งบอกความสามารถในการทำงานของ CYP จากผลของลักษณะทางพันธุกรรมที่ผิดปกติ

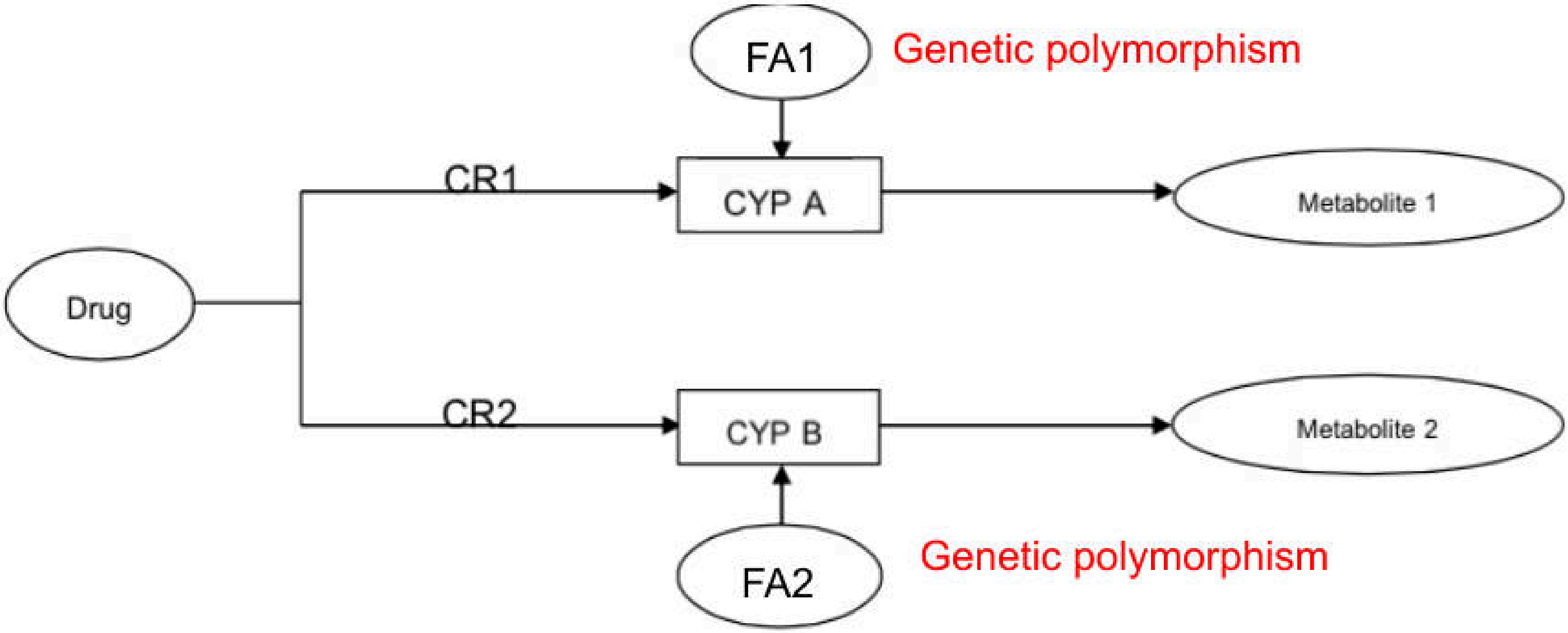
PK-DGIs


Poor metabolizers (PMs)
or
Intermediate metabolizers (IMs)
or
Extensive metabolizers (EMs)
or
Rapid metabolizers (RMs)
Ultrarapid metabolizers (UMs)
or

Frequency of Genetic Polymorphisms Producing Slow Metabolism in Some DME and Representative Substrates*

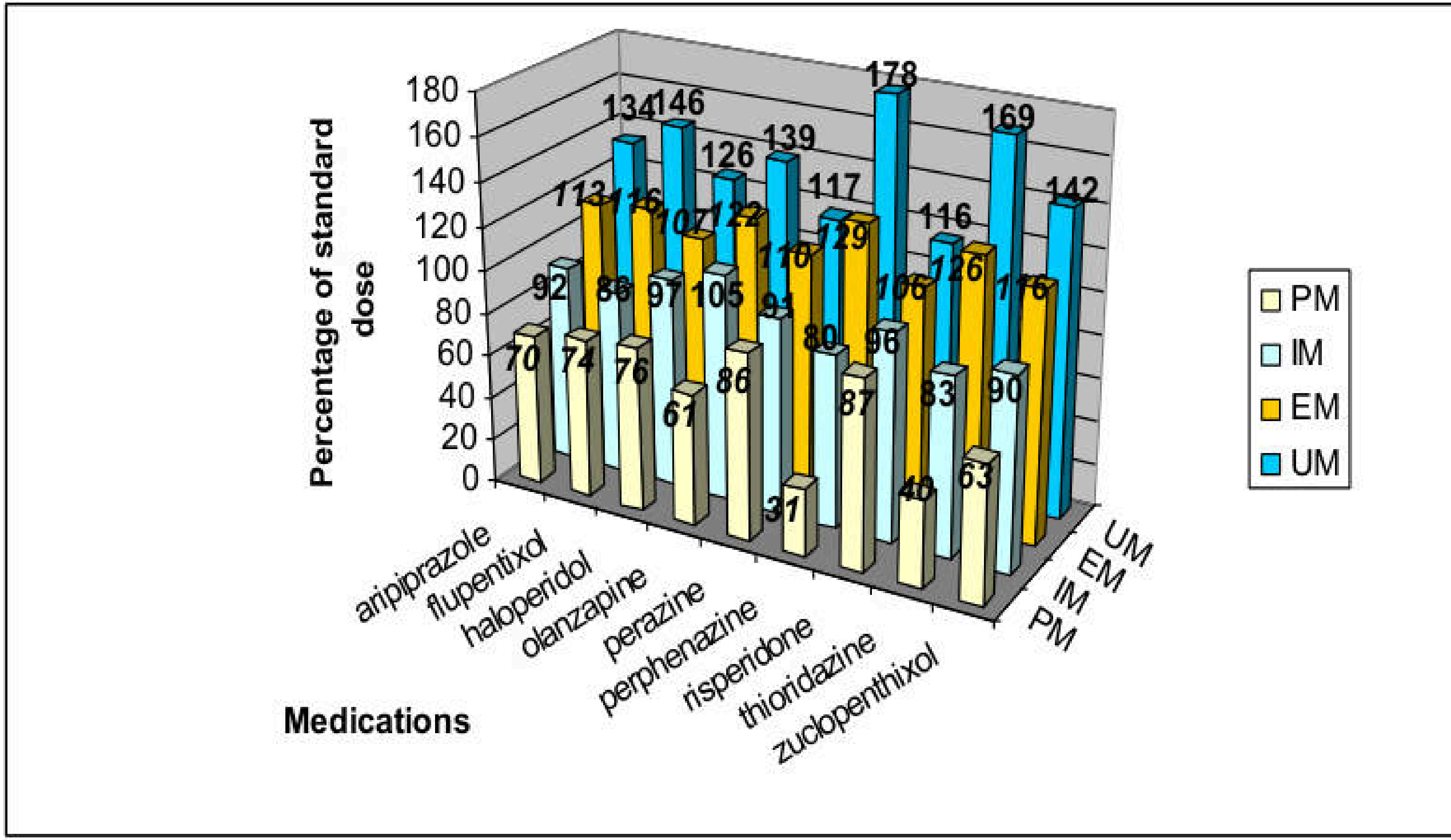
Enzyme	Frequency of Poor Metabolizer Status	Drug Substrates†
Phase I Reactions		
CYP2D6	5%–10% Caucasians 3.8% African Americans 0.9% Asians 1% Arabs	Bufurolol, codeine, dextromethorphan, encainide, flecainide, metoprolol, nortriptyline, timolol
CYP2C9	1%–3% Caucasians	Celecoxib, fluvastatin, glyburide, S-ibuprofen, tolbutamide, phenytoin, S-warfarin
CYP2C19	3%–5% Caucasians 16% Asians	Diazepam, lansoprazole, omeprazole, pantoprazole.
Butylcholinesterase	Several abnormal genes; most common disorder 1 in 2500	Succinylcholine
Phase II Reactions		
Thiopurine S-methyltransferase	0.3% Caucasians 0.04% Asians	Azathioprine, mercaptopurine
N-Acetyltransferase (NAT2)	60% Caucasians, African Americans 10%–20% Asians	Amrinone, hydralazine, isoniazid, phenelzine, aminosalicylic acid
Uridine diphosphate glucuronyl transferase	11% Caucasians 1%–3% Asians	Irinotecan

PK-Drug-Gene-Gene Interaction (DGGI) Model



FA = fractional activity คือ ค่าที่บ่งบอกความสามารถในการทำงานของ CYP จากผลของลักษณะทางพันธุกรรมที่ผิดปกติ

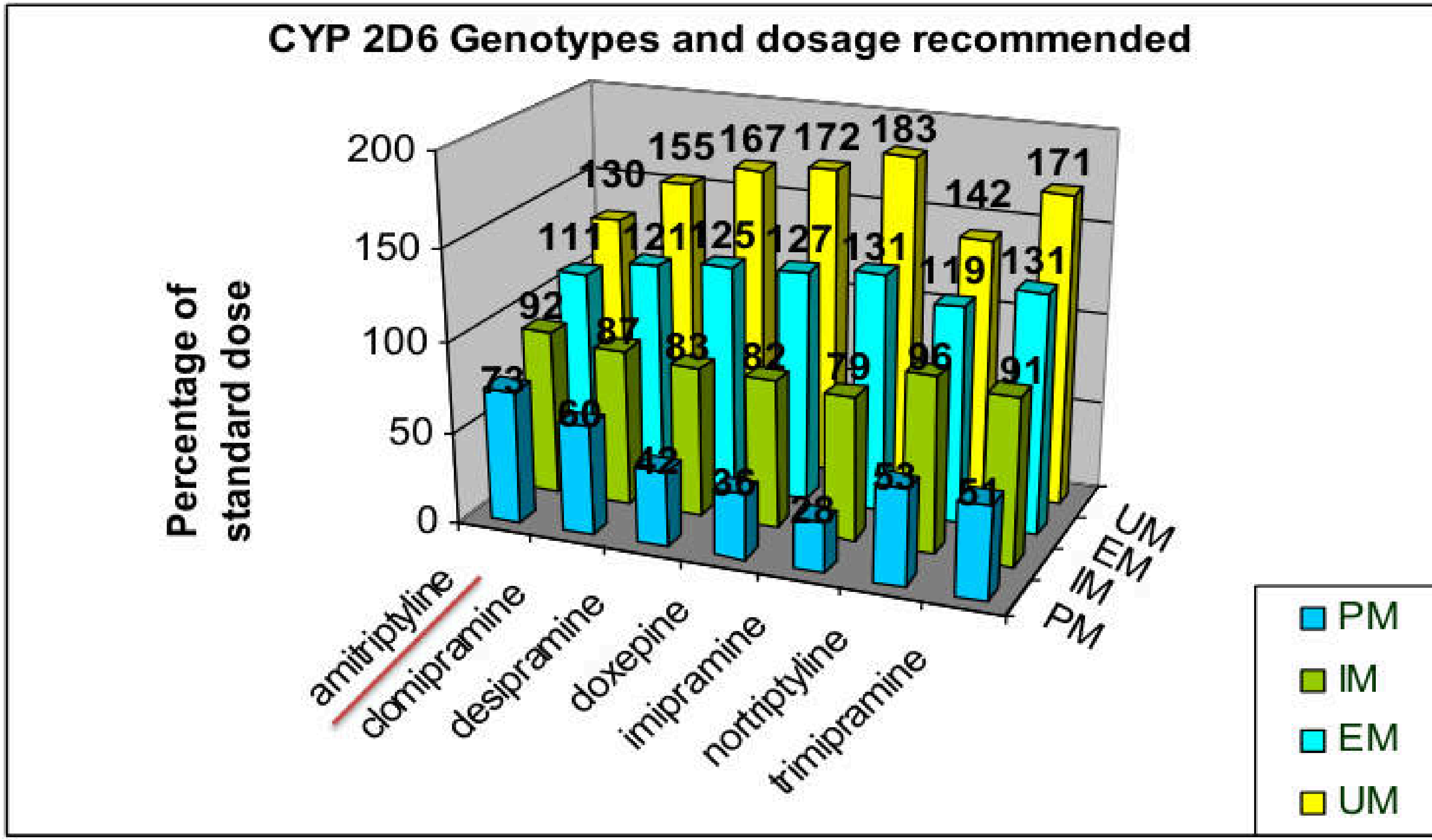
PK-DGI - Efficacy: 2D6 AND ANTIPSYCHOTICS



Antipsychotic dose adjustments are recommended for 2D6 PM and UM

Kirchheiner J, Nickchen K, Bauer M, et el. Mol Psychiatry 2004 May; 9 (5):442-73.

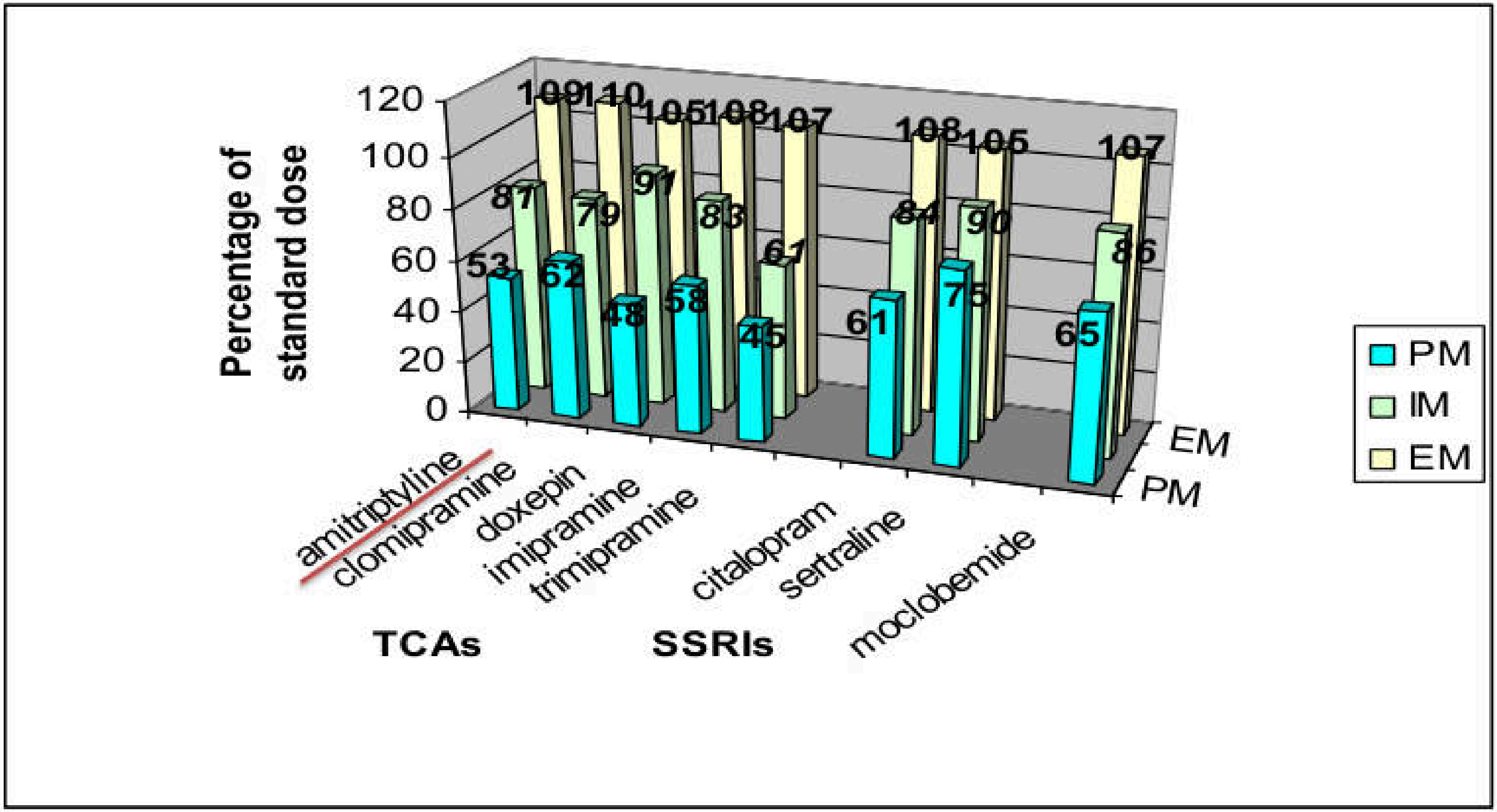
PK-DGI - Efficacy: CYP2D6 AND TCAs



TCA dose adjustments are recommended for 2D6 PM and UM.

Kirchheiner J, Nickchen K, Bauer M, et el. Mol Psychiatry 2004 May; 9 (5):442-73.

PK-DGI – Efficacy: 2C19 AND ANTIDEPRESSANTS



Recommendation: 2C19 PM: 60% of standard doses

2C19 EM: 110% of standard doses

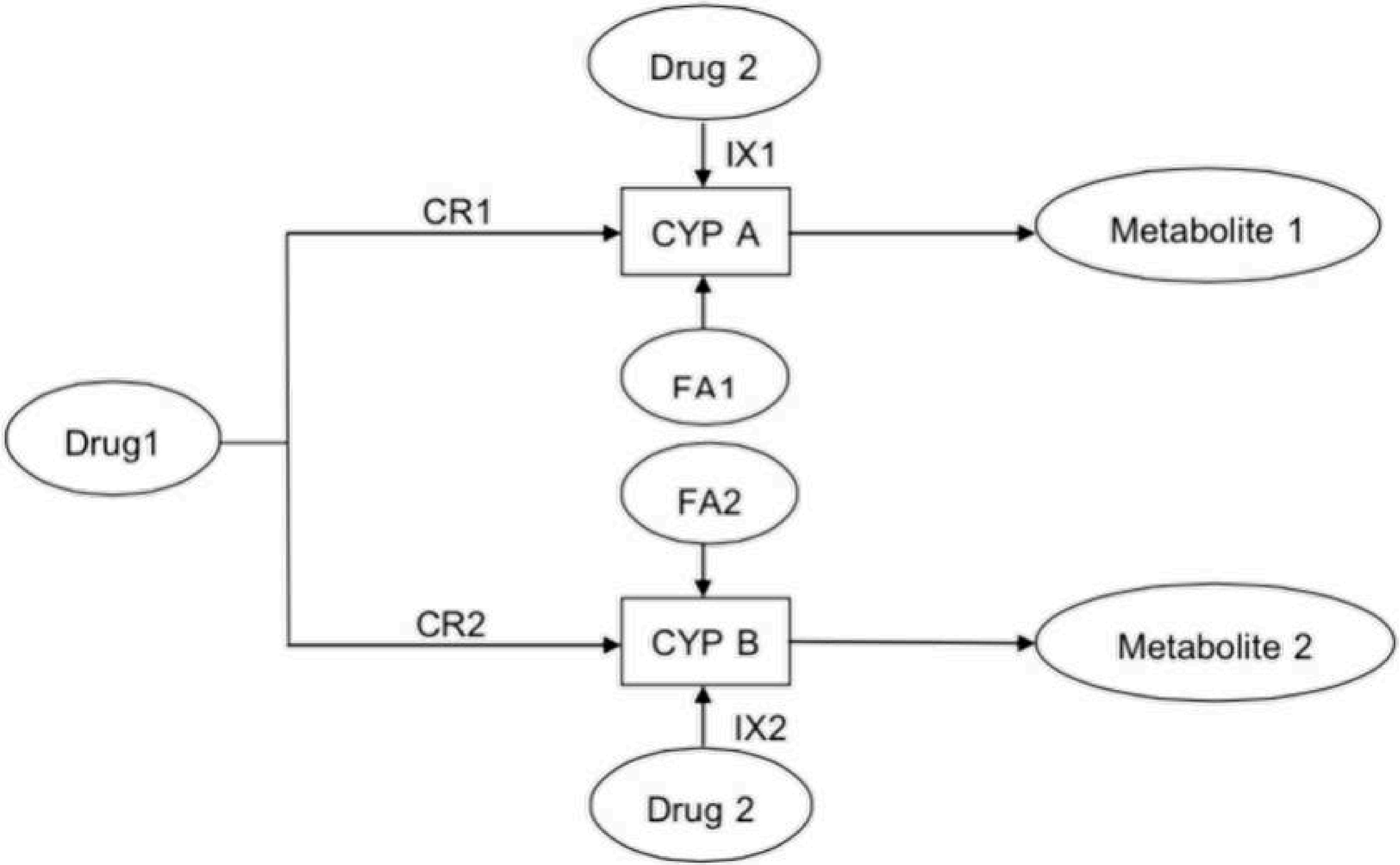
Kirchheiner J, Nickchen K, Bauer M, et el. Mol Psychiatry 2004 May; 9 (5):442-73.

PK-DGGI: Dosing Recommendation For TCA

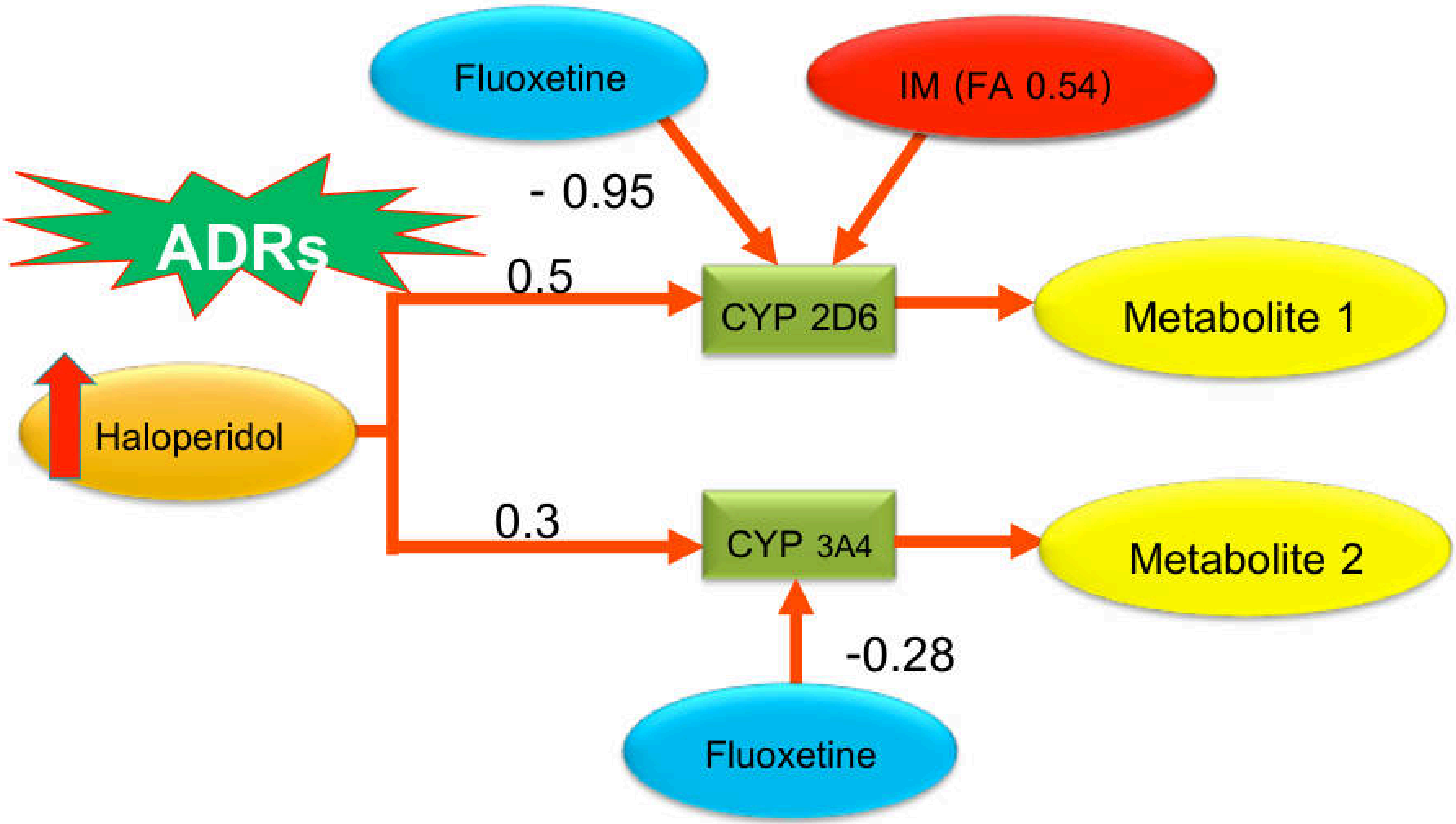
Supplemental Table S19. Dosing Recommendations for Amitriptyline Based on Both CYP2D6 and CYP2C19 Phenotype: [1,2,3,4]

Phenotype	CYP2D6 Ultrarapid metabolizer	CYP2D6 Extensive metabolizer	CYP2D6 Intermediate metabolizer	CYP2D6 Poor metabolizer
CYP2C19 Ultrarapid metabolizer	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustment.	Consider alternative drug not metabolized by CYP2C19. If a tricyclic is warranted, utilize therapeutic drug monitoring to guide dose adjustments.	Consider alternative drug not metabolized by CYP2C19. If a tricyclic is warranted, utilize therapeutic drug monitoring to guide dose adjustments.	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustment.
CYP2C19 Extensive metabolizer	Avoid tricyclic use. If a tricyclic is warranted consider increasing the starting dose. [1] Utilize therapeutic drug monitoring to guide dose adjustments.	Initiate therapy with recommended starting dose. [1]	Consider 25% reduction of recommended starting dose. [1] Utilize therapeutic drug monitoring to guide dose adjustments.	Avoid tricyclic use. If a tricyclic is warranted consider 50% reduction of recommended starting dose. [1] Utilize therapeutic drug monitoring to guide dose adjustment.
CYP2C19 Intermediate metabolizer	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustments.	Initiate therapy with recommended starting dose. [1]	Consider 25% reduction of recommended starting dose. [1] Utilize therapeutic drug monitoring to guide dose adjustments.	Avoid tricyclic use. If a tricyclic is warranted consider 50% reduction of recommended starting dose a Utilize therapeutic drug monitoring to guide dose adjustment.
CYP2C19 Poor metabolizer	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustments.	Consider 50% reduction of recommended starting dose. [1] Utilize therapeutic drug monitoring to guide dose adjustments.	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustments.	Avoid tricyclic use. If a tricyclic is warranted utilize therapeutic drug monitoring to guide dose adjustments.

PK-Drug-Drug Gene Gene Interaction (DDGGI) Model



PK-DDGI -Type A ADR: Haloperidol

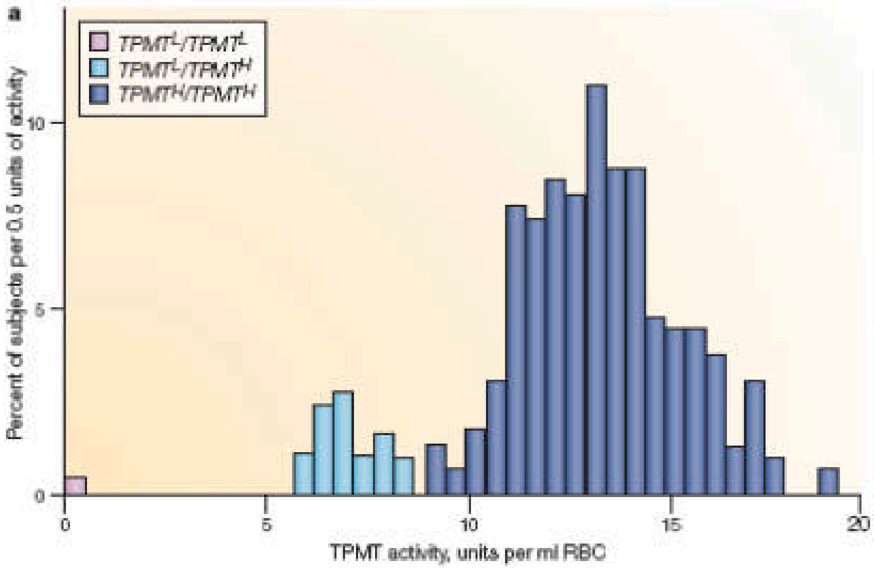


PK DGI – Type A ADR: Thiopurine drugs

- Azathioprine : immunosuppresant
- 6-Mercaptopurine: anticancer
- 6-Thioguanine: anticancer

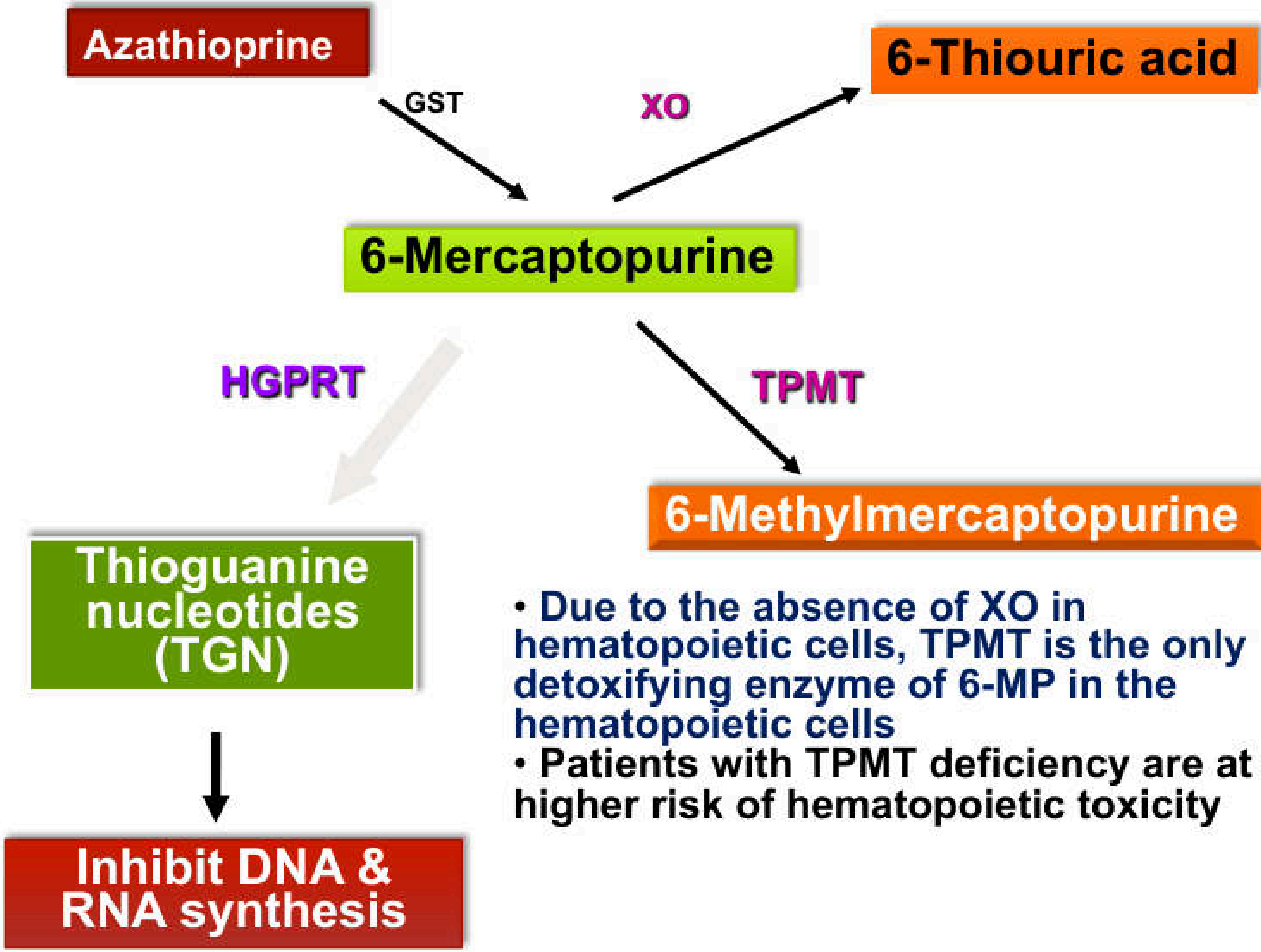
Mainly metabolized by thiopurine S-methyltransfearse (TPMT)

Inter-individual difference in TPMT activity



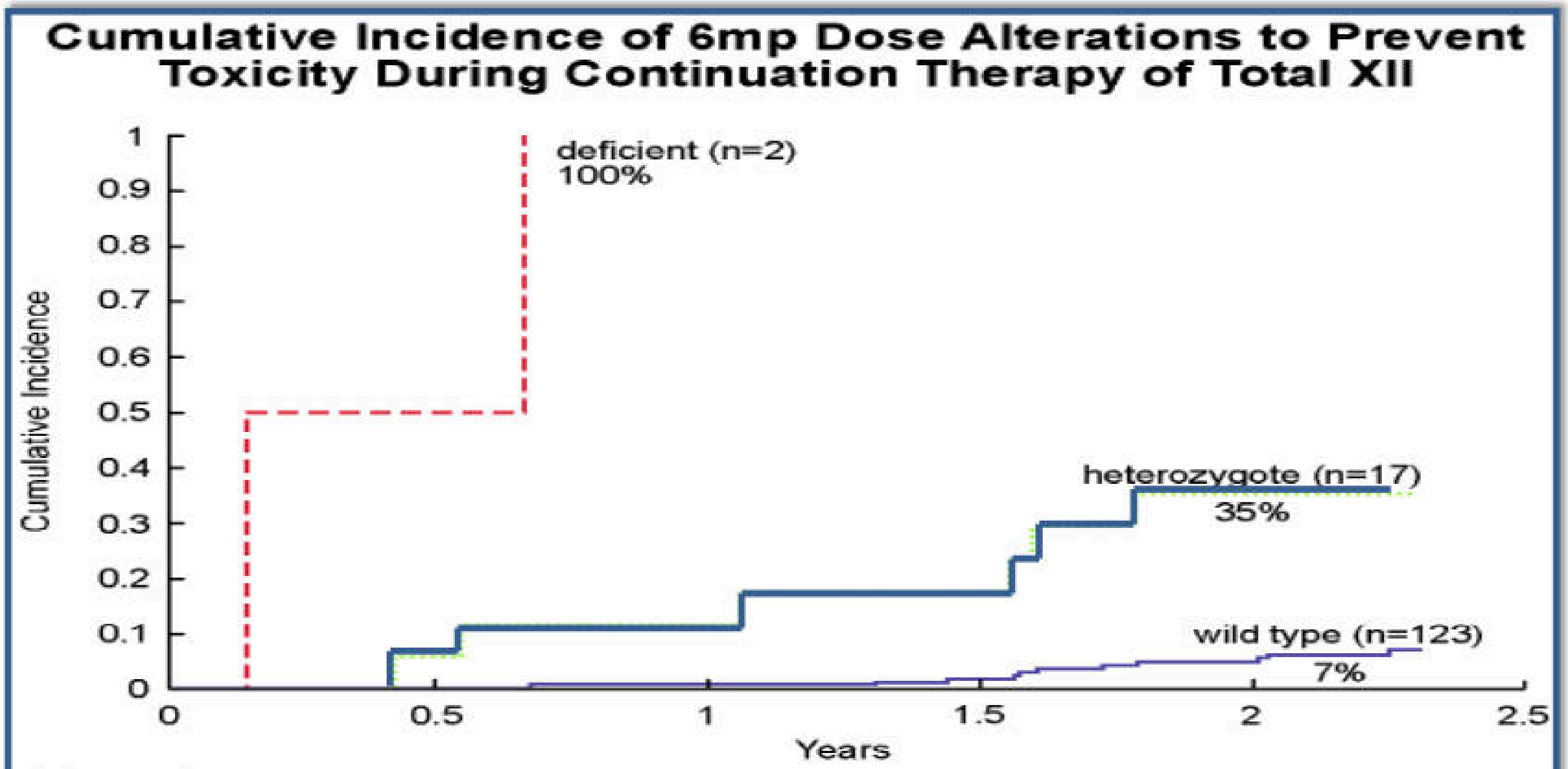
For Thai :10% for IM
0.5% for PM
(Tassaneeyakul et al., Br J Clin Pharmacol 2004; 58:66-70)

Veinshilboum and Sladex, Am J Hum Gen , 32:651-62, 1980



- Due to the absence of XO in hematopoietic cells, TPMT is the only detoxifying enzyme of 6-MP in the hematopoietic cells
- Patients with TPMT deficiency are at a higher risk of hematopoietic toxicity

Cumulative incidence of hematotoxicity during 6-MP therapy



Relling et al
J Natl Cancer Inst. 91(23):2001-8, 1999

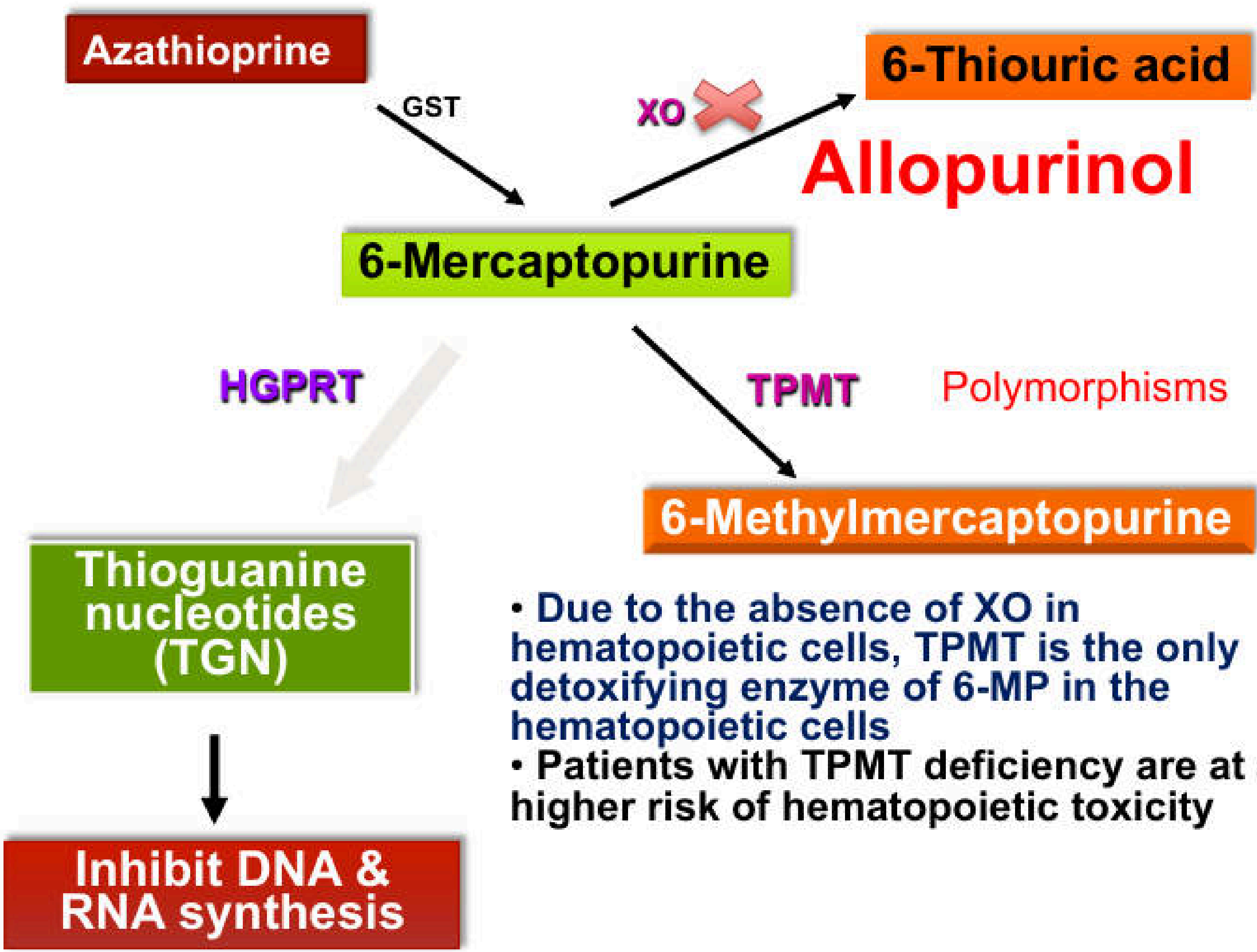
Risk of AZA-induced hematoxicity in kidney transplant patients

TPMT genotype	Total number of subject	Non-hematotoxicity (N)	Hematotoxicity (N)	Crude OR (95% CI)	Adjusted OR (95% CI)
**1/*1*	130	114 (87.69%)	16 (12.31%)	1.0	1.0
**1/*3C*	9	3 (33.33%)	6 (66.67%)	14.25 (3.24-62.69)	14.18 (3.07-65.40)

Risk of hematotoxicity in kidney transplant pts with heterozygous of *TPMT*3C* was 14 fold higher than the wild-type patients

Vannaprasaht S, et al. Clin Ther. 2009;31:1524-33

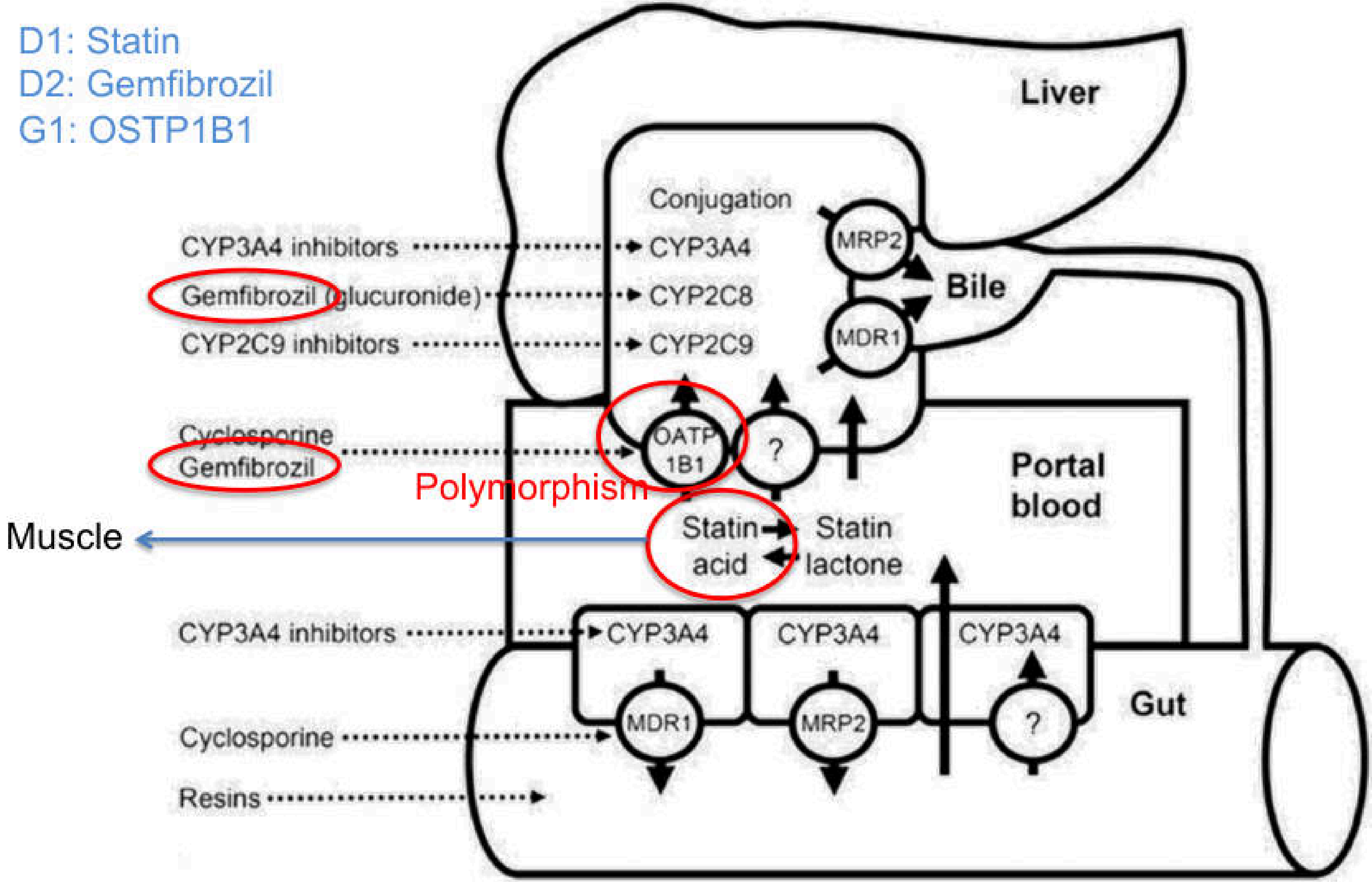
PK-DDGI – Type A ADR: Thiopurine drugs



- Due to the absence of XO in hematopoietic cells, TPMT is the only detoxifying enzyme of 6-MP in the hematopoietic cells
- Patients with TPMT deficiency are at a higher risk of hematopoietic toxicity

PK DDGI – Type A ADR: Simvastatin induced Myopathy

D1: Statin
D2: Gemfibrozil
G1: OSTP1B1

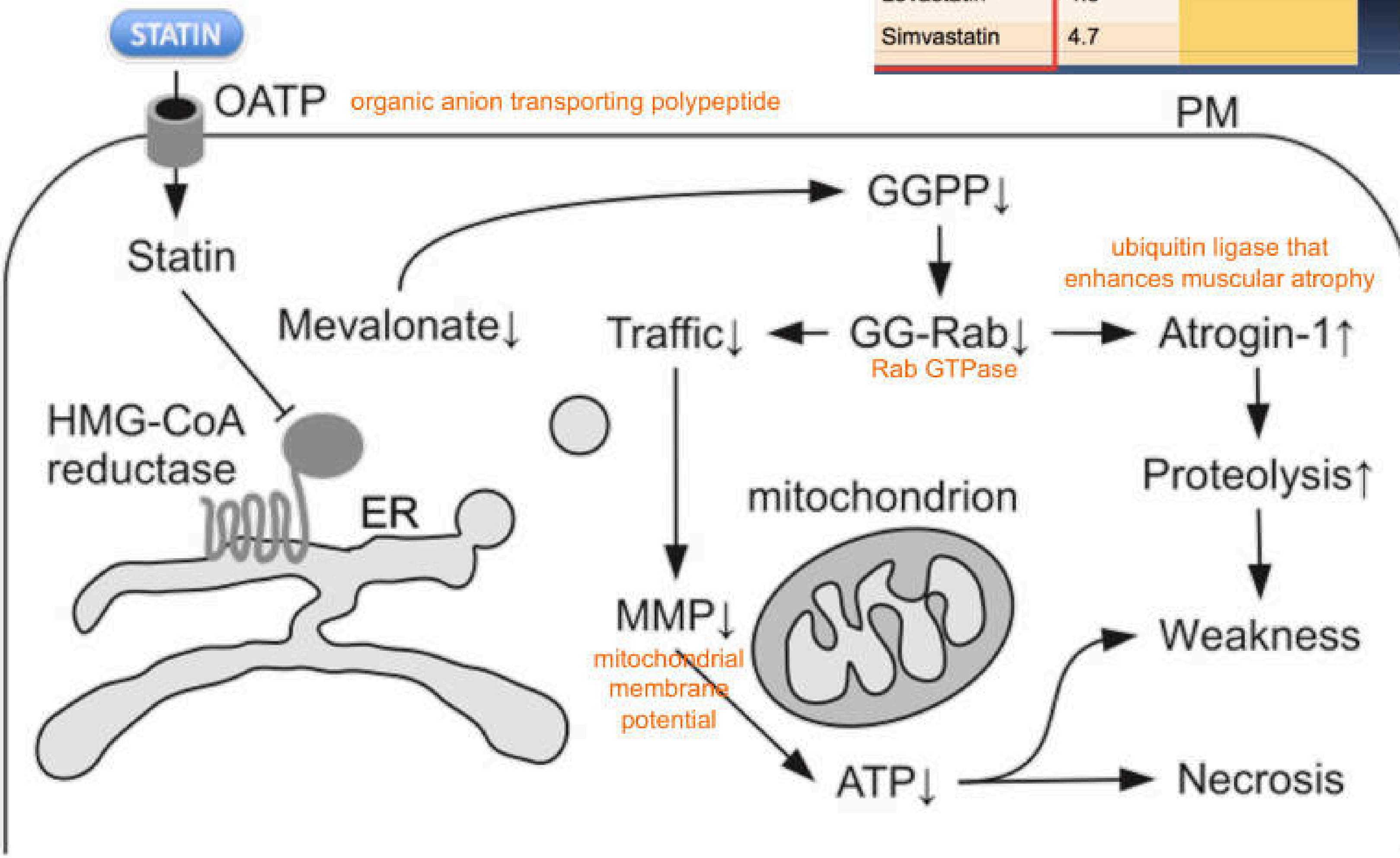


Clinical Pharmacology & Therapeutics (2006) 80, 565–581

Proposed Mechanisms of Statin-Associated Myopathy

Statin	Log P	
Rosuvastatin	-0.3	Hydrophilicity
Pravastatin	-0.2	
Fluvastatin	3.2	Lipophilicity
Pitavastatin	3.5	
Atorvastatin	4.1	
Lovastatin	4.3	
Simvastatin	4.7	

Less uptake into muscle cell



J Pharmacol Sci **123**, 289 – 294 (2013)

ThaiSNP2

Polymorphisms Information :

Query SNP Id	:	rs4149056
Co-located SNP Id	:	rs4149056, ssj0003182
Class	:	snp
Sequences	:	GGTTGTTTAAAGGAATCTGGGTCATAC [C/T] GTTCATGGGTAATATGCTTCGTGGAAT Show/hide full length of flanking sequ
Position	:	at NT_009714.16 : 14090523 Chr12
Validation Status	:	Validated
Polymorphism in gene	:	SLCO1B1 (GeneID:10599) mRNA : NI
Source	:	
Graphical view	:	

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GENE:
SLCO1B1
solute carrier organic anion transporter family, member 1B1

Clinical PGx | PGx Research | Overview | VIP | Haplotypes | Pathways | Is Related To | Publications | Downloads/LinkOuts

Dosing Guidelines (1) | Drug Labels (0) | Clinical Annotations (31) | Genetic Tests (3)

CPIC Dosing Guideline for simvastatin and SLCO1B1 — *last updated 06/30/2014*

Summary

The FDA recommends against 80mg daily simvastatin dosage. In patients with the C allele at SLCO1B1 rs4149056, there are modest increases in myopathy risk even at lower simvastatin doses (40mg daily); if optimal efficacy is not achieved with a lower dose, alternate agents should be considered.

Find genotype-based dosing recommendation

Pick rs4149056 alleles: C | C

Phenotype (Genotype)

Low activity

Implications

High myopathy risk

Recommendations (Strength: Strong)

FDA recommends against 80 mg. Prescribe a lower dose or consider an alternative statin; consider routine creatine kinase (CK) surveillance.

At the time of the development of this recommendation, there are no data available on the possible role of SLCO1B1 in simvastatin-induced myopathies in pediatric patient populations; however, there is no reason to suspect that SLCO1B1 variant alleles would affect simvastatin hepatic uptake differently in children compared to adults. Please see below for full details of these guidelines, with supporting evidence and disclaimers.

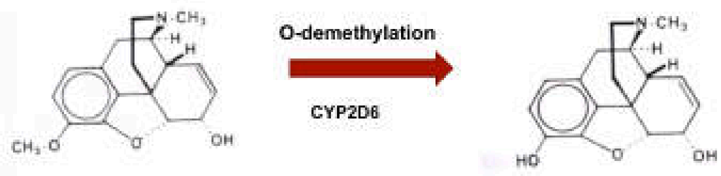
Risk allele C: 13% in Thais

Variation Summary :

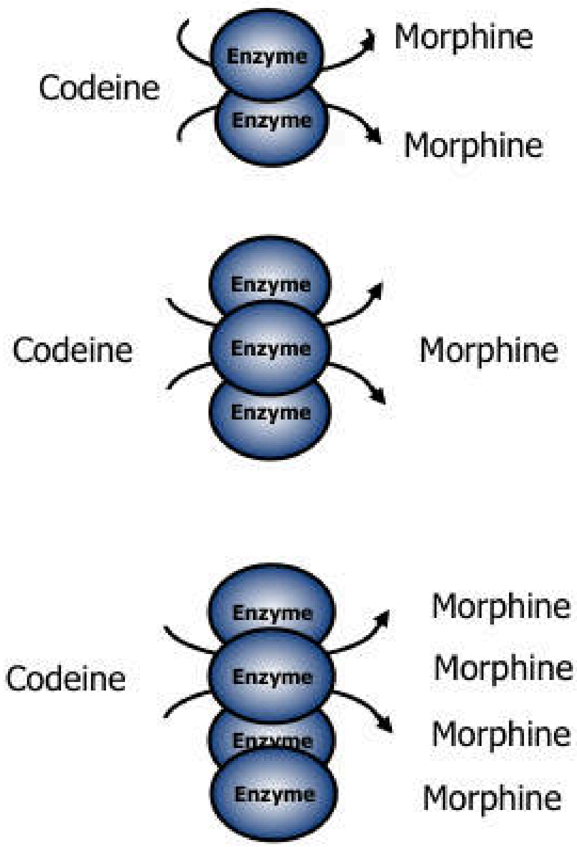
Source	Population	Sample size	Allele Freq	Genotype Freq	Observed Heterozygosity	Expected Heterozygosity
dbSNP	---	980	C [0.099] / T [0.901]	---	---	0.180
HapMap	ASW	166	T [0.946] / C [0.054]	TT [0.904] / TC [0.084] / CC [0.012]	0.084	0.103
HapMap	CEU	330	T [0.867] / C [0.133]	TT [0.752] / TC [0.230] / CC [0.018]	0.230	0.231
HapMap	CHB	168	T [0.851] / C [0.149]	TT [0.714] / TC [0.274] / CC [0.012]	0.274	0.253
HapMap	CHD	170	T [0.876] / C [0.124]	TT [0.776] / TC [0.200] / CC [0.024]	0.200	0.216
HapMap	GIH	176	T [0.977] / C [0.023]	TT [0.955] / TC [0.045] / CC [0.000]	0.045	0.044
HapMap	JPT	172	T [0.890] / C [0.110]	TT [0.802] / TC [0.174] / CC [0.023]	0.174	0.197
HapMap	LWK	180	T [0.983] / C [0.017]	TT [0.967] / TC [0.033] / CC [0.000]	0.033	0.033
HapMap	MEX	154	T [0.922] / C [0.078]	TT [0.844] / TC [0.156] / CC [0.000]	0.156	0.144
HapMap	MKK	342	T [0.889] / C [0.111]	TT [0.789] / TC [0.199] / CC [0.012]	0.199	0.198
HapMap	TSI	176	T [0.784] / C [0.216]	TT [0.602] / TC [0.364] / CC [0.034]	0.364	0.339
HapMap	YRI	334	T [0.988] / C [0.012]	TT [0.976] / TC [0.024] / CC [0.000]	0.024	0.024
Depression Contro	Thai	376	T [0.870] / C [0.130]	---	---	0.226

: dbSNP, : HapMap, : JSNP, : PASNP, : ThaiSNP1, : ThaiSNP2, : Thai depression-control, : Thai drug related genes (TCELS)

PK DGI – Type A ADR: Codeine-Respiratory Depression



Codeine **Morphine**



Extensive Metabolizer: Normal CYP2D6 activity

Response well to codeine
(Right amount of morphine)

Poor Metabolizer: Low CYP2D6 activity

No analgesic effect
(Too small amount of morphine)

Ultrarapid Metabolizer: High CYP2D6 activity

Respiratory suppression
(Too much amount of morphine)

Life-threatening complication after cough suppression therapy with codeine

- 62 yr man with pneumonia treated with codeine (25 mg tid) for cough
- 4 days after drug administration , the pt's consciousness rapidly deteriorated,

and he became unresponsive.

- At the time of the pt's coma,
  - plasma morphine was 80 μg/L (normal 1-4 μg/L)
  - morphine–3–glucuronide was 580 μg/L (normal 8-70 μg/L)
  - morphine–6–glucuronide was 136 μg/L (normal 1-13 μg/L)
- CYP2D6 genotyping : ultra rapid metabolizer

NEJM2004;351:2827-31

Case Report

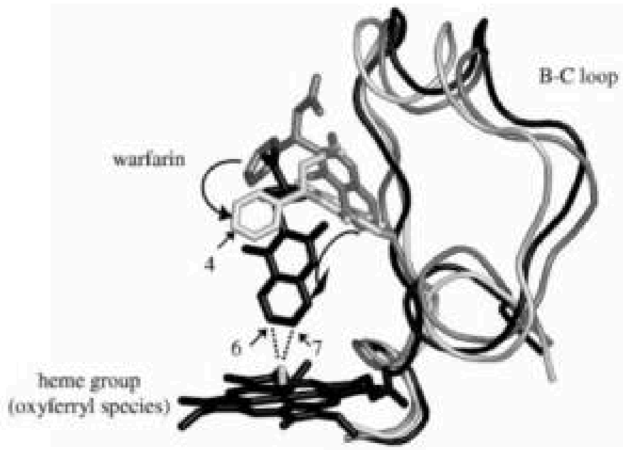
Lancet 2006; 368: 704

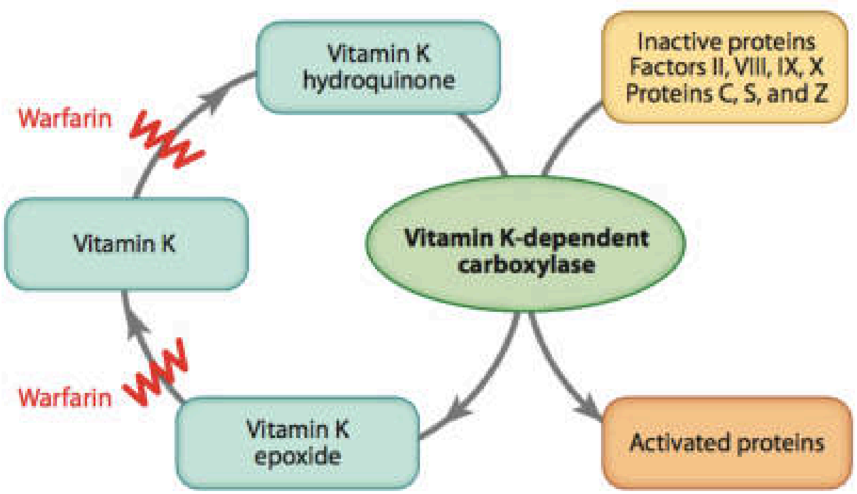
Pharmacogenetics of morphine poisoning in a breastfed neonate of a codeine-prescribed mother

Gideon Koren, James Cairns, David Chitayat, Andrea Gaedigk, Steven J Leeder

- **A women had been prescribed a combination of codeine 30 mg and paracetamol 500 mg (2 tab bid for 2 wk) after giving birth for episiotomy pain.**
- **She was breastfeeding the infant and on day 7 after birth, her infant showed intermittent periods of difficulty in breast feeding and lethargy.**
- **On day 12, he had grey skin and his milk intake had fallen and found dead on day 13**
- **Postmortem analysis showed no anatomical anomalies. Blood concentration of morphine in the neonate was 70 ng/mL (generally should be about 0–2 ng/mL)**
- **Because of poor neonatal feeding, she stored milk on day 10, which was later assayed and found morphine concentration of 87 ng/mL (typical range of codeine concentration in milk is 2–20 ng/mL at doses of 60 mg every 6 h)**
- **Mother's CYP2D6 genotype : *CYP2D6*2 x 2* (Ultra rapid metaboizer)**

PK, PD, DGGI type A ADR: Warfarin induced Bleeding





- Anticoagulant used to prevent thrombotic events
- Metabolized by mainly by CYP2C9

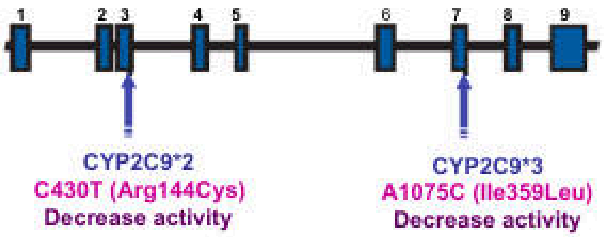
Mechanism of action:

Inhibit the C1 subunit of the vitamin K epoxide reductase (VKORC1) complex, thereby reducing the regeneration of reduced vitamin K which necessary for clotting factors

Narrow therapeutic window

Major ADR : Bleeding in internal organs

CYP2C9 Allele distribution in Thais



CYP2C9 polymorphism in a Thai population

Clearance of warfarin in patients with CYP2C9*2 or CYP2C9*3 are slower than homozygous CYP2C9*1 patients

VKORC1 haplotype distribution in Thais

VKORC1: reduce vitamin K 2,3-epoxide to reduced vitamin K which required for blood coagulation

A Haplotype : −1639A, 1173T
B Haplotype : −1639G, 1173C

VKORC1 polymorphism in a Thai population

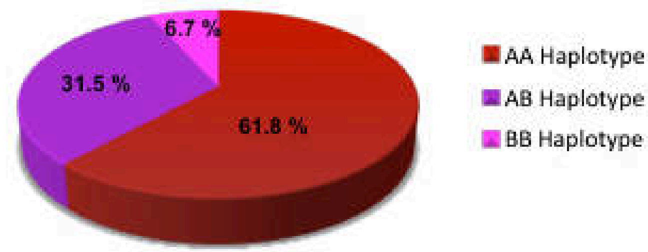


Table 5: Range of Expected Therapeutic Warfarin Doses Based on CYP2C9 and VKORC1 Genotypes†

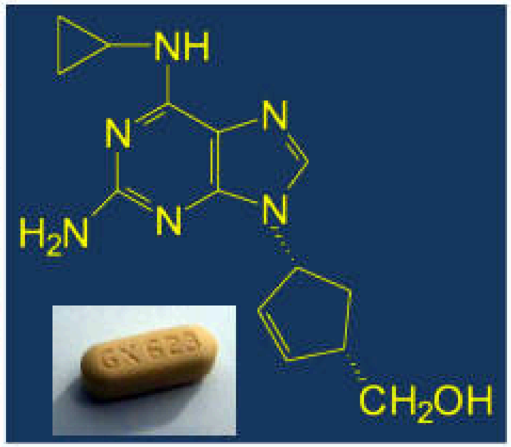
VKORC1 -1639	CYP2C9					
	*1/*1	*1/*2	*1/*3	*2/*2	*2/*3	*3/*3
GG	5-7 mg	5-7 mg	3-4 mg	3-4 mg	3-4 mg	0.5-2 mg
AG	5-7 mg	3-4 mg	3-4 mg	3-4 mg	0.5-2 mg	0.5-2 mg
AA	3-4 mg	3-4 mg	0.5-2 mg	0.5-2 mg	0.5-2 mg	0.5-2 mg

If the patient's CYP2C9 and VKORC1 genotypes are not known, the initial dose of COUMADIN is usually 2 to 5 mg per day

Off-target DGI- type B ADR: Abacavir Hypersensitivity

- Nucleoside analogue
- Reverse transcriptase inhibitor

Clinical phenotype



Causal chemical

- Hypersensitivity 5%
- Fever, skin rash, GI symptoms, eosinophilia within 6 weeks

Clinical genotype

Association with HLA-B*5701

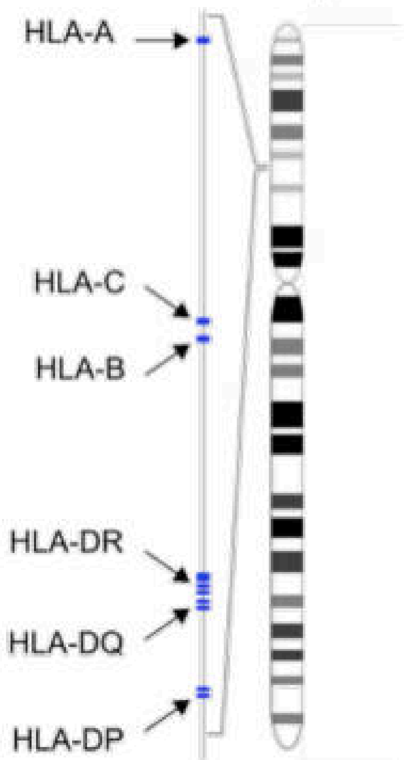
Cost-effectiveness analysis of HLA *B*5701* genotyping in preventing abacavir hypersensitivity
Dyfrig A. Hughes[a], F. Javier Vilar[b], Charlotte C. Ward[a], Ana Alfirevic[a], B. Kevin Park[a] and Munir Pirmohamed[a]

Incidence before and after testing for HLA-B*5701

Country	Pre testing	Post testing	Reference
Australia	7%	<1%	Rauch et al, 2006
France	12%	0%	Zucman et al, 2007
UK (London)	7.8%	2%	Waters et al, 2007

HLA polymorphism in population

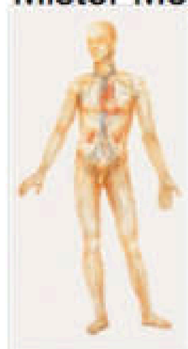
HLA
MHC Complex



Chr. 6

ca 14 HLA alleles / individual
ca. 7800 HLA alleles in human population
some alleles are frequent, some are rare

Mister Meyer



HLA

- A*02:01, *A* 31:01*
- *B*10:02; B*57:01*
- C* 02:01; *C*06:01*
- DR* B1 01:01; *04:02*
- *DR** B5 01:01
- *DP* 04:04; 08:01*
- *DQ* 01:05; 05:01*

Risk for Carbamazepine hypersensitivity

Abacavir hypersensitivity

Mister Wang

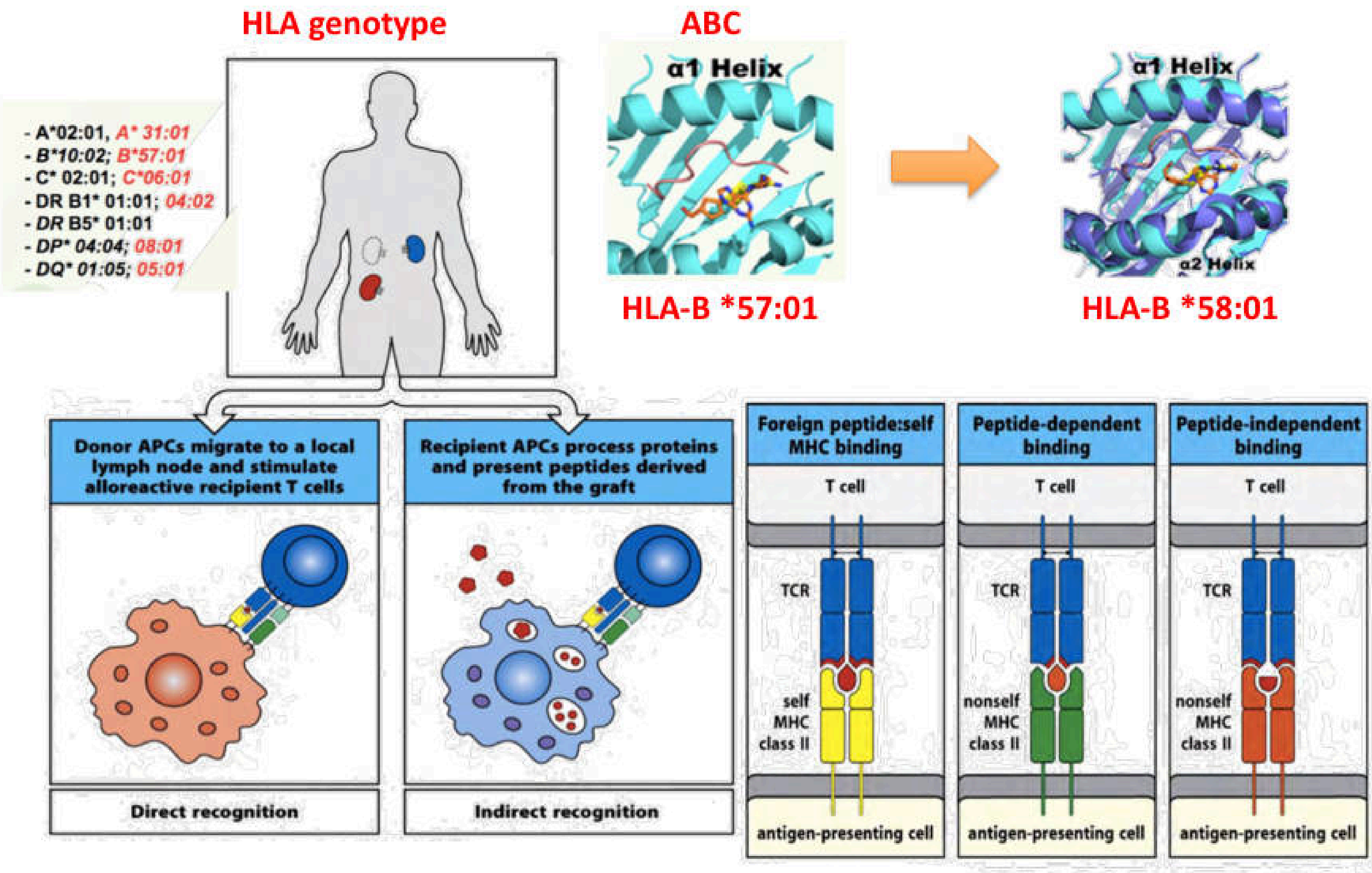
HLA

- A*02:01, *A* 32:01*
- *B* 15:02; B*58:01*
- C* 0201; *C*06:01*
- DR* B1-01:01; *04:02*
- *DR** B5-01:01
- *DP* 03:04; 07:01*
- *DQ* 02:04; 03:02*

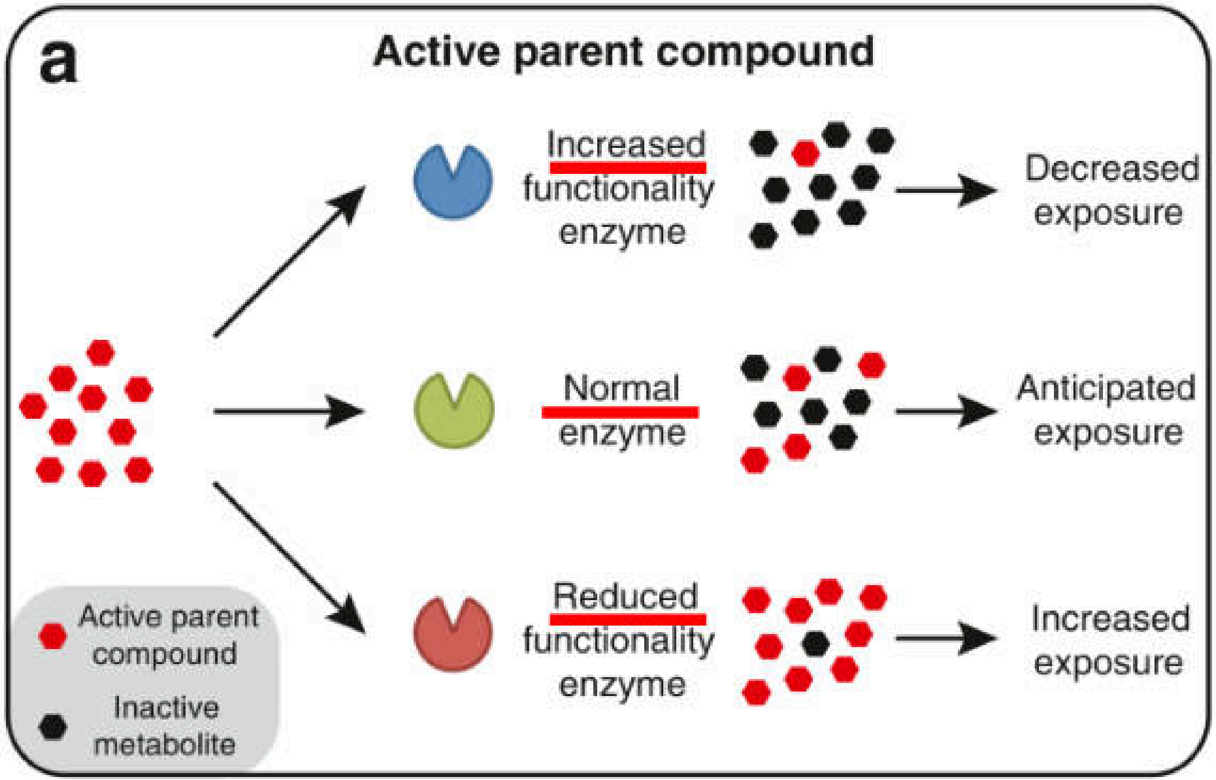
Risk for Allopurinol hypersensitivity

Risk for Carbamazepine hypersensitivity

Abacavir - Allo-reactivity (as GVHD)



ยาที่ต้องถูกเปลี่ยนแปลงด้วยเอนไซม์ให้หมดฤทธิ์



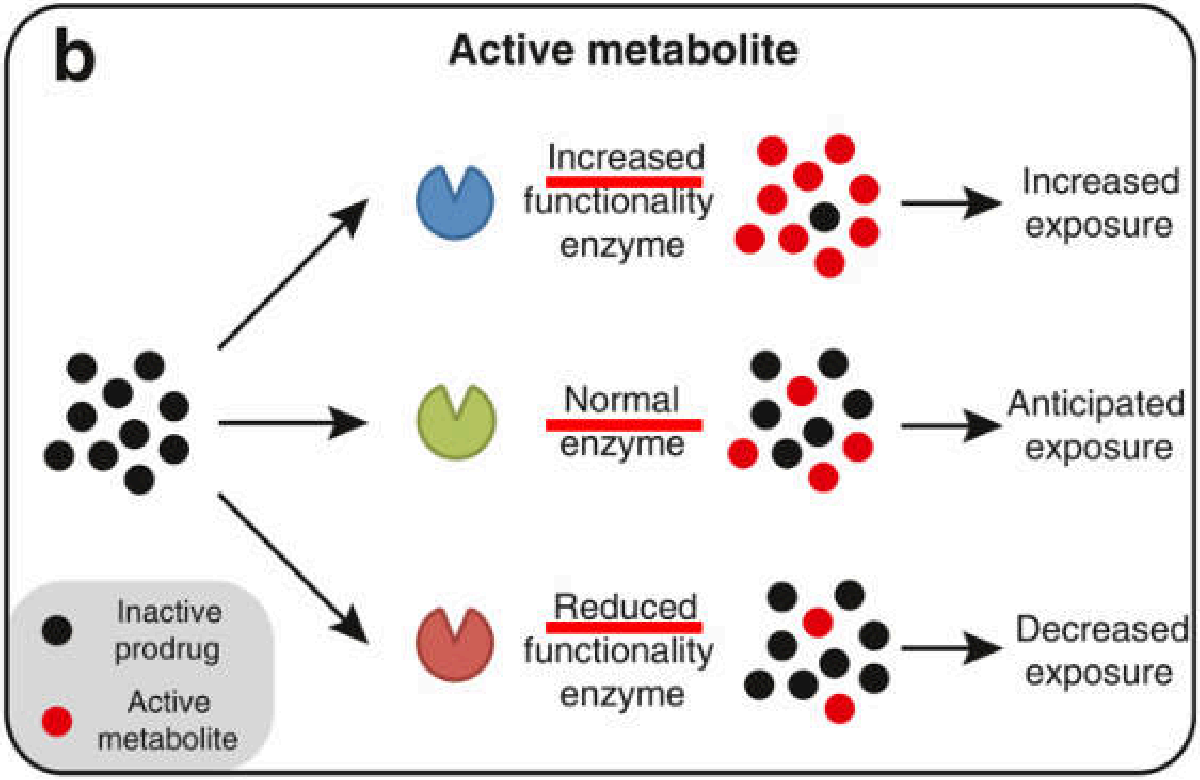
Pharmacokinetics

เพิ่มขนาดยา

ขนาดยาเท่าเดิม

ลดขนาดยาลง

ยาที่ต้องถูกเปลี่ยนแปลงด้วยเอนไซม์ให้มีฤทธิ์ก่อนการรักษา



Pharmacokinetics

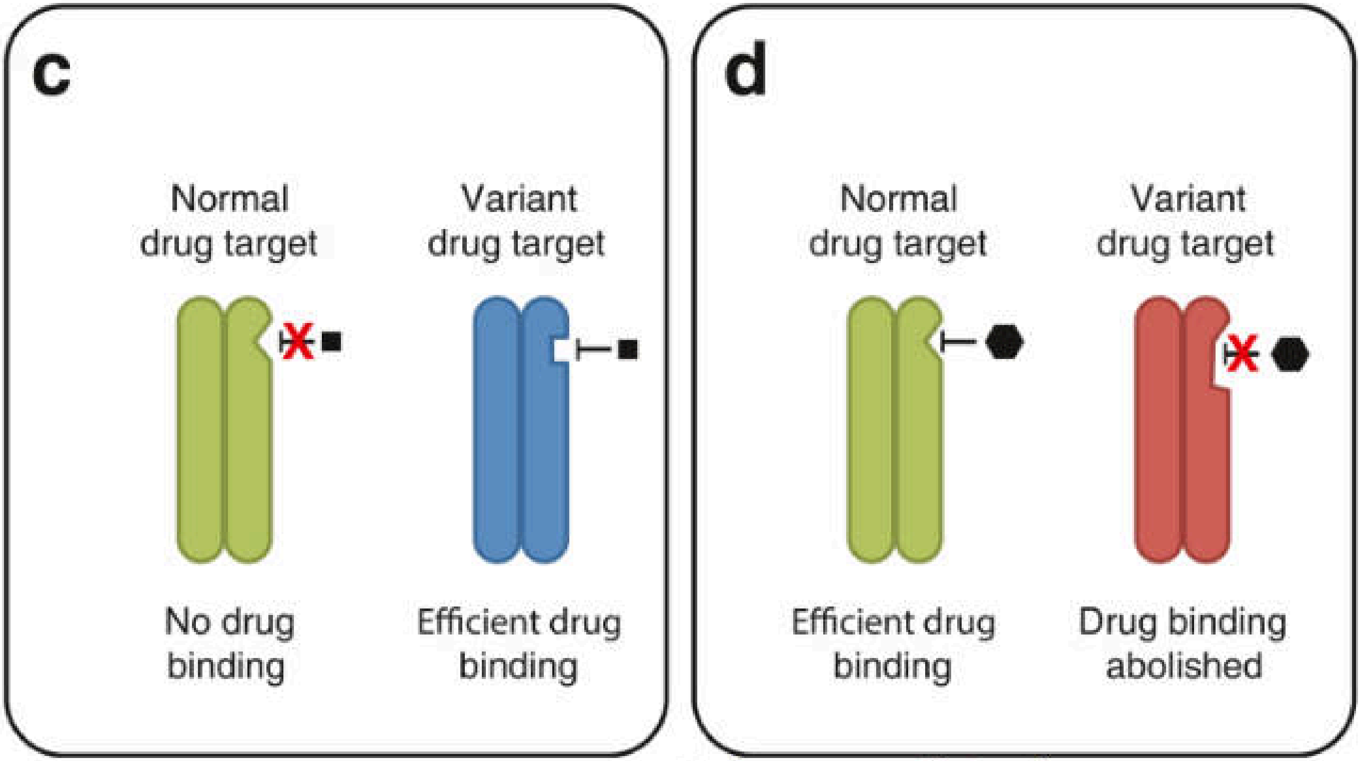
ลดขนาดยาลง

ขนาดยาเท่าเดิม

เพิ่มขนาดยา

ยาจับเป้าหมายแตกต่างกัน

Pharmacodynamics

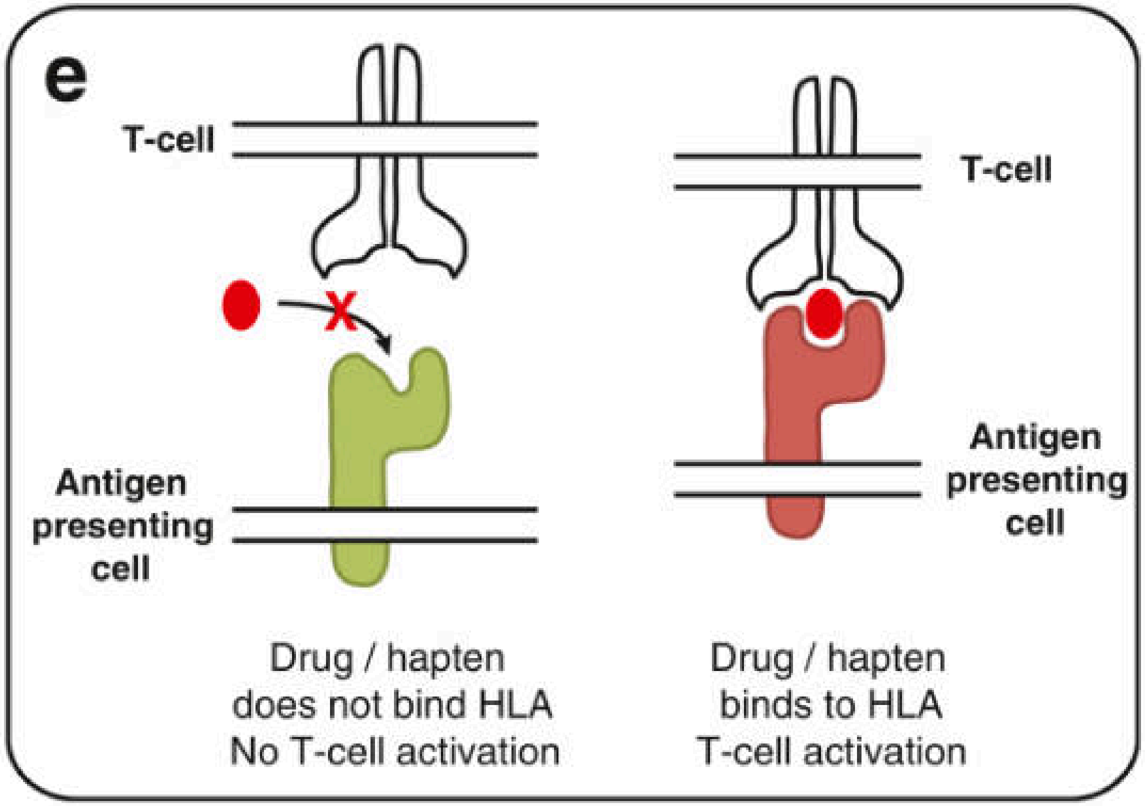


เลือกคนไข้ที่มีเป้าหมายจำเพาะ

ไม่ใช้ยาในคนไข้ที่มีเป้าหมายของยาผิดปกติ

ยาจับผิดเป้าหมาย

Off-target



เปลี่ยนยาในคนไข้ที่ยาจับเป้าหมายของการแพ้ยาได้